Automated Construction of Multi-Element MEAM Interatomic Potentials

DFT-Informed Neural Network Optimization

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Motivation

- Industrial alloys (steels, Al alloys) are selected by their **elastic properties**: stiffness vs. elongation.
- Experimental measurement of elastic properties is **expensive and time-consuming**.
- Molecular simulations can predict elastic properties computationally—but they require accurate **interatomic potentials**.
- The Modified Embedded Atom Method (MEAM) is widely used for metals and alloys Baskes (1992); Lee and Baskes (2001).

Goal

Build an automated pipeline to generate MEAM potentials for *arbitrary* multi-element alloys.

The Problem: Fragmented MEAM Libraries

Published MEAM files cover only **subsets** of elements. No single file spans all elements needed for industrial alloys (e.g. Al 7075: Al, Zn, Mg, Cu, Cr, Mn, Fe, Si, Ti).

Library File	Al	Co	Cr	Cu	Fe	Mg	Mn	Mo	Ni	Si	Ti	Zn
AlMgZn	✓					✓						✓
AlSiCuMgFe	✓			✓	✓	✓				✓		
CuMo				✓				✓				
FeMnNiTiCuCrCoAl	✓	✓	✓	✓	✓		✓		✓		✓	

Cross-interaction parameters between files are **undefined** $\Rightarrow$ cannot simulate combined compositions.

Research Question

Can density functional theory reference calculations with neural-network surrogate optimization produce multi-element MEAM interatomic potentials that accurately predict the elastic properties of arbitrary metallic alloy compositions, even when the constituent elements do not share a common published MEAM parameterization?

Pipeline Overview

Python	LAMMPS (Python API)	Quantum Espresso + ASE	TensorFlow + LAMMPS
Stage 1 Config Generation	**Stage 2 MD Simulation**	**Stage 3 DFT Reference**	**Stage 4 NN Surrogate Optimization**
Random alloy compositions	E, ν per config → results.json	a_0, E_{coh}, B, C_{ij} → MEAM init	Inverse optimize MEAM params
✓ Done	✓ Done	✓ Done	In Progress

Stage 1: Configuration Generation — Python

Random alloy compositions from 12 MEAM-available elements.

Properties:

- 10^{-4} composition precision
- Al dominant when present
- Fractions sum to exactly 1.0
- 5800+ configs generated

Example: AL 7075 simplified

```
{
  "composition": {
    "Al": 0.91,
    "Mg": 0.029,
    "Zn": 0.061
  },
  "box_size_m": 4e-09,
  "temperature": 0.1,
  "total_steps": 1000
}
```

Stage 2: MD Simulation — LAMMPS (Python API) + OVITO

Static deformation method to compute elastic properties:

1. **Relax** — anisotropic pressure minimization (box + atoms)
2. **Baseline** — record stress $\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$
3. **Strain** — apply $\pm\delta$ ($\delta = 0.001$) along x, y, z with re-minimization
4. **Extract** — C_{11} from axial, C_{12} from transverse stress derivatives
5. **Average** — 3 C_{11} and 6 C_{12} samples across directions
6. **Compute:**

$$E = \frac{(C_{11} - C_{12})(C_{11} + 2C_{12})}{C_{11} + C_{12}}, \qquad \nu = \frac{C_{12}}{C_{11} + C_{12}}$$

Configs with $\nu < 0$ are discarded as unphysical.

Stage 2: MD Results

402 simulations completed across random and named alloy compositions.

Alloy Config	E **(GPa)**	ν
AL 2024 (Al–Cu–Mn)	66.7	0.354
AL 2219 (Al–Cu)	74.0	0.351
AL 5052 (Al–Mg)	73.1	0.343
AL 7075 (Al–Mg–Zn)	67.1	0.346
Random multi-component examples:		
$Al_{50}Mn_{41}Ni_9$	77.6	0.329
$Al_{50}Ni_{21}Mn_{15}Cu_8Cr_6$	217.2	0.158
$Al_{50}Ti_{38}Co_{12}$	226.6	0.273

Young's modulus range: $\sim$40–230 GPa. Results stored in `results.json` for downstream NN training.

Stage 3: DFT Reference Calculations — Quantum Espresso + ASE

Quantum Espresso computes reference data for each element:

Per-element properties:

- Equation of state (7 strain points)
- Birch–Murnaghan fit $\rightarrow a_0$, E_{coh}, B
- Elastic constants C_{11}, C_{12}
- Isolated atom energy for true E_{coh}

Binary pairs:

- Formation energy E_{form} for all $\binom{N}{2}$ pairs
- 11 elements $\rightarrow$ 36 binary pairs computed

Pseudopotentials auto-downloaded from QE library if missing.

	a_0 (Å)	E_{coh}	B
Al	4.05	3.61	77.5
Cu	3.64	3.86	138.5
Fe	2.81	6.88	420.2
Ti	2.91	6.29	114.5
Cr	2.85	8.67	209.4

E_{coh} in eV, B in GPa

Stage 3: DFT → MEAM — Single-Element Parameters

Direct mappings

$$\texttt{esub} \leftarrow E_{\mathrm{coh}} \qquad \text{(cohesive / sublimation energy)}$$

$$\texttt{alat} \leftarrow a_0 \qquad \text{(equilibrium lattice constant)}$$

Rose equation shape parameter

$$\texttt{alpha} = \sqrt{\frac{9\,B\,\Omega}{E_{\mathrm{coh}}}}$$

where B is the bulk modulus (converted to eV/Å^3) and Ω is the atomic volume:

FCC: $\Omega = a^3/4$ BCC: $\Omega = a^3/2$

HCP: $\Omega \approx a^3\sqrt{2}$ Diamond: $\Omega = a^3/8$

Stage 3: DFT → MEAM — Cross-Interaction Parameters

For each binary pair (i, j), DFT formation energies set the cross-terms:

Binary pair parameters

$$E_c(i,j) = \frac{\mathtt{esub}_i + \mathtt{esub}_j}{2} + E_{\mathrm{form}}(i,j)$$

$$r_e(i,j) = \frac{d_{\mathrm{nn}}(i) + d_{\mathrm{nn}}(j)}{2}$$

$$\alpha(i,j) = \frac{\alpha_i + \alpha_j}{2}$$

where d_{nn} is the nearest-neighbor distance derived from DFT lattice constants.

36 binary pairs computed from 11 elements — covers all $\binom{11}{2} = 55$ combinations where DFT data is available.

Stage 3: Example — Aluminum → MEAM

DFT Output	MEAM Parameter
$a_0 = 4.047\,\text{Å}$	`alat` = 4.047
$E_{\text{coh}} = 3.609\,\text{eV}$	`esub` = 3.609
$B = 77.5\,\text{GPa}$	`alpha` = 4.61

`alpha` derivation for Al (FCC)

1. Unit conversion: $B = 77.5\ \text{GPa} \times 0.00624 = 0.4836\ \text{eV}/\text{Å}^3$
2. Atomic volume: $\Omega_{\text{FCC}} = a_0^3/4 = 4.047^3/4 = 16.58\ \text{Å}^3$
3. Rose equation: $\alpha = \sqrt{9 \times 0.4836 \times 16.58\ /\ 3.609} = \sqrt{19.99} = 4.61$

Stage 3: MEAM File Merging — Python

When no single library file covers all selected elements:

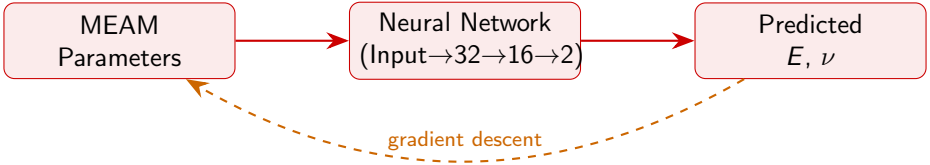
1. **Scan** all `EAM/library_*.meam` files
2. **Find** minimal covering set of files for the target elements
3. **Merge** single-element entries into a unified library file
4. **Fill** missing cross-interaction parameters from DFT results
5. **Write** merged `library_<elements>.meam` + `<elements>.meam`

Example: Al–Cu–Zn–Mg–Fe–Cr–Ti

- Al, Cu, Fe, Cr, Ti from `FeMnNiTiCuCrCoAl`
- Mg, Zn from `AlMgZn`
- Cross-terms (e.g. Cr–Mg, Ti–Zn) computed from DFT formation energies

Stage 4: NN Surrogate Optimization — TensorFlow + LAMMPS

Status: In Progress



1. **Sample:** Perturb DFT-initialized MEAM parameters
2. **Evaluate:** Run LAMMPS for each sample across target compositions
3. **Train:** NN learns mapping `MEAM_params` $\rightarrow (E, \nu)$
4. **Optimize:** Back-propagate through trained NN to find optimal parameters
5. **Validate:** Final LAMMPS evaluation with optimized parameters

Expected Outcomes

1. **Unified MEAM potentials** — Complete library files covering all selected elements with consistent cross-interaction terms.
2. **DFT-quality initialization** — Elemental properties and binary formation energies matching published DFT values, providing a physically valid optimization starting point.
3. **NN surrogate accuracy** — A trained model that predicts elastic response of MEAM parameter sets, enabling rapid inverse optimization without repeated LAMMPS evaluations.

References I

Baskes, M. I. (1992). Modified embedded-atom potentials for cubic materials and impurities. *Physical Review B*, 46(5):2727–2742.

Lee, B.-J. and Baskes, M. I. (2001). Second nearest-neighbor modified embedded-atom-method potential. *Physical Review B*, 64(18):184102.